

5.6.23. Fibonacci Variation: A single pair of rabbits (male and female) is born at the beginning of a year. Assume the following conditions:

(1) Rabbit pairs are not fertile during their first two months of life, but thereafter give birth to three new male/female pairs at the end of every month.

(2) No rabbits die.

a. Let s_n = the number of pairs of rabbits alive at the end of month n , for each integer $n \geq 1$, and let $s^0 = 1$. Find a recurrence relation for $s^0, s^1, s^2, \dots$. Justify your answer.

b. Compute s_0, s_1, s_2, s_3, s_4 , and s_5 .

c. How many rabbits will there be at the end of the year?

a. Let s_n = the number of pairs of rabbits alive at the end of month n , for each integer $n \geq 1$, and let $s^0 = 1$. Find a recurrence relation for $s^0, s^1, s^2, \dots$. Justify your answer.

Alright, so what do we know?

We know that s_n is the number of pairs of rabbits alive at the end of month n . So n is the month.

Furthermore, we know that $s_0 = 1$, which means that we start with one pair of rabbits.

Rabbits do not give birth during their first two months of life, so we may assume:

$$s_0 = 1$$

$$s_1 = 1$$

$$s_2 = 1$$

Now, the pair gives birth three new pairs. Therefore:

$$s_3 = 1 + 3 = 4$$

We have four pairs in total, whereas the first pair will keep giving birth to three new pairs, and the current pairs will not start giving birth until two months after they were born.

Hence:

$$s_4 = 4 + 3 = 7$$

$$s_5 = 7 + 3 = 10$$

By keeping track of the pairs giving birth from s^6 through s^8 , we find that:

$$s_6 = 10 + 3 + 3 = 22$$

$$s_7 = 22 + 3 + 3 + 3 = 43$$

$$s_8 = 43 + 3 + 3 + 3 + 3 = 73$$

Now, if we put them all together, we might be able to spot a recurrence relation:

$$s_0 = 1$$

$$s_1 = 1$$

$$s_2 = 1$$

$$s_3 = 1 + 3 = 4$$

$$\begin{aligned}
s_4 &= 4 + 3 &&= 7 \\
s_5 &= 7 + 3 &&= 10 \\
s_6 &= 10 + 3 + 3 + 3 + 3 &&= 22 \\
s_7 &= 22 + 3_2 + 3_2 + 3 &&= 43 \\
s_8 &= 43 + 3_2 + 3_2 + 3_2 + 3 &&= 73
\end{aligned}$$

Which can be written as:

$$\begin{aligned}
s_0 &= 1 \\
s_1 &= 1 \\
s_2 &= 1 \\
s_3 &= 1 + 3 &&= 4 \\
s_4 &= 4 + 3 &&= 7 \\
s_5 &= 7 + 3 &&= 10 \\
s_6 &= 10 + 3(4) &&= 22 \\
s_7 &= 22 + 3(7) &&= 43 \\
s_8 &= 43 + 3(10) &&= 73
\end{aligned}$$

Which in turn can be written as:

$$\begin{aligned}
s_0 &= 1 \\
s_1 &= 1 \\
s_2 &= 1 \\
s_3 &= s_2 + 3 &&= 4 \\
s_4 &= s_3 + 3 &&= 7 \\
s_5 &= s_4 + 3 &&= 10 \\
s_6 &= s_5 + 3s_3 &&= 22 \\
s_7 &= s_6 + 3s_4 &&= 43 \\
s_8 &= s_7 + 3s_5 &&= 73
\end{aligned}$$

But this does not give us the supposed pattern. Let's look closer at what happens with the subscripts:

$$\begin{aligned}
s_6 &= s_5 + 3s_3 \\
&\quad 6 \rightarrow 5 \text{ and } 6 \rightarrow 3 \\
s_7 &= s_6 + 3s_4 \\
&\quad 7 \rightarrow 6 \text{ and } 7 \rightarrow 4 \\
s_8 &= s_7 + 3s_5 \\
&\quad 8 \rightarrow 7 \text{ and } 8 \rightarrow 5
\end{aligned}$$

Which can be denoted as:

$$s_n = s_{n-1} + 3s_{n-3}$$

Suppose our guess is this pattern as denoted above. Let's apply it to our known range and

see whether it holds:

$$s_0 = 1$$

$$s_1 = 1$$

$$s_2 = 1$$

$$s_3 = s_2 + 3s_0$$

$$s_4 = s_3 + 3s_1$$

$$s_5 = s_4 + 3s_2$$

$$s_6 = s_5 + 3s_3$$

$$s_7 = s_6 + 3s_4$$

$$s_8 = s_7 + 3s_5$$

Since $s_2 + 3s_0$ is $1 + 3 * 1 = 4$, and $s_3 + 3s_1$ is $4 + 3 * 1 = 7$, we suppose to guess that the pattern is $s_n = s_{n-1} + 3s_{n-3}$, which is what was to be shown.

Solution: $s_n = s_{n-1} + 3s_{n-3}$ is the recurrence relation as motivated above.

b. Compute s_0, s_1, s_2, s_3, s_4 , and s_5 .

Suppose $s_n = s_{n-1} + 3s_{n-3}$, then:

$$s_0 = 1 \quad = 1$$

$$s_1 = 1 \quad = 1$$

$$s_2 = 1 \quad = 1$$

$$s_3 = s_2 + 3s_0 \quad = 4$$

$$s_4 = s_3 + 3s_1 \quad = 7$$

$$s_5 = s_4 + 3s_2 \quad = 10$$

Solution: Compute $s_0 = 1, s_1 = 1, s_2 = 1, s_3 = 4, s_4 = 7$, and $s_5 = 10$.

c. How many rabbits will there be at the end of the year?

Suppose $s_n = s_{n-1} + 3s_{n-3}$, then:

Month	Calc.	Sum
$s_0 = 1$	$= 1$	$= 1$
$s_1 = 1$	$= 1$	$= 1$
$s_2 = 1$	$= 1$	$= 1$
$s_3 = s_2 + 3s_0$	$= 1 + 3(1)$	$= 4$
$s_4 = s_3 + 3s_1$	$= 4 + 3(1)$	$= 7$
$s_5 = s_4 + 3s_2$	$= 7 + 3(1)$	$= 10$
$s_6 = s_5 + 3s_3$	$= 10 + 3(4)$	$= 22$
$s_7 = s_6 + 3s_4$	$= 22 + 3(7)$	$= 43$
$s_8 = s_7 + 3s_5$	$= 43 + 3(10)$	$= 73$
$s_9 = s_8 + 3s_6$	$= 73 + 3(22)$	$= 139$
$s_{10} = s_9 + 3s_7$	$= 139 + 3(43)$	$= 268$
$s_{11} = s_{10} + 3s_8$	$= 268 + 3(73)$	$= 487$
$s_{12} = s_{11} + 3s_9$	$= 487 + 3(139)$	$= 904$

Since we know that there're 904 pairs at the end of the year:

Therefore, $904 * 2 = 1808$ rabbits.

Solution: There're 1808 rabbits at the end of the year.